$$\Rightarrow \alpha = \frac{1}{137}$$

(b) Using fine structure constant, the velocity of electron can be written as

$$\upsilon_n = \frac{\alpha c Z}{n}$$

For hydrogen atom ($Z = 1$) the velocity of electron in n^{th} orbit is

$$\upsilon_n = \frac{c}{137}\frac{1}{n} = (2.19\times10^6)\frac{1}{n}\ \text{ms}^{-1}$$

For the first orbit (ground state), the velocity of electron is

$$\upsilon_1 = 2.19\times10^6\,\text{ms}^{-1}$$

For the second orbit (first excited state), the velocity of electron is

$$\upsilon_2 = 1.095\times10^6\,\text{ms}^{-1}$$

For the third orbit (second excited state), the velocity of electron is

$$\upsilon_3 = 0.73\times10^6\,\text{ms}^{-1}$$

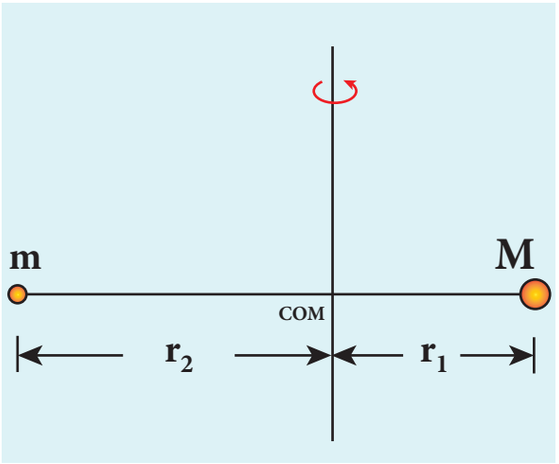
Here, $v_1 > v_2 > v_3$

EXAMPLE 9.4

The Bohr atom model is derived with the assumption that the nucleus of the atom is stationary and only electrons revolve around the nucleus. Suppose the nucleus is also in motion, then calculate the energy of this new system.

Solution

Let the mass of the electron be m and mass of the nucleus be M. Since there is no external force acting on the system, the centre of mass of hydrogen atom remains at rest. Hence, both nucleus and electron move about the centre of mass as shown in figure.



Let V be the velocity of the nuclear motion and v be the velocity of electron motion. Since the total linear momentum of the system is zero,

$$-m\upsilon + M\upsilon = 0 \text{ or}$$

$$MV = m\upsilon = p$$

$$\vec{p}_e + \vec{p}_n = \vec{0} \text{ or}$$

$$\left|\vec{p}_e\right| = \left|\vec{p}_n\right| = p$$

Hence, the kinetic energy of the system is

$$KE = \frac{p_n^2}{2M} + \frac{p_e^2}{2m} = \frac{p^2}{2}\left(\frac{1}{M} + \frac{1}{m}\right)$$

Let $\frac{1}{M} + \frac{1}{m} = \frac{1}{\mu_m}$. Here the reduced mass is, $\mu_m = \frac{mM}{M+m}$

Therefore, the kinetic energy of the system now is $KE = \frac{p^2}{2\mu_m}$

Since the potential energy of the system is same, the total energy of the hydrogen can be expressed by replacing mass by reduced mass, which is

$$E_n = -\frac{\mu_m e^4}{8\varepsilon_0^2 h^2}\frac{1}{n^2}$$

Since the nucleus is very heavy compared to the electron, the reduced mass is closer to the mass of the electron.

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